

Toward Outdoor Population Presence Monitoring With Mobile Network Data and Satellite Imagery

Marta Alonso Tubía , Miguel Baena Botana , An Vo Quang, Ana Burgin, and Oliva Garcia Cantú-Ros 

Abstract—Dynamic population mapping has become crucial for capturing real-time human movement and behavior, beyond traditional population mapping relying on census data. Differentiating indoor and outdoor activity enhances accuracy for smart city planning, emergency response, public health, or emerging technologies like Innovative Air Mobility, where pedestrian data informs safer, less disruptive flight planning. Data passively collected from mobile networks have proven to be highly effective in accurately capturing population presence and mobility patterns. By enhancing this rich data source with GPS data for spatial accuracy and validating the results with satellite imagery of detected pedestrians, we provide a procedure for indoor and outdoor population detection. The results show agreement between both methodologies. Despite some limitations related to GPS data biases and pedestrian detection issues caused by urban furniture and shadows, the procedure demonstrates strong potential to capture people's movements, which could ultimately enable near real-time monitoring of population presence on the streets.

Index Terms—Autonomous aerial vehicles, convolutional neural networks (CNNs), Earth observation, global positioning system, location awareness, mobile computing, mobile location, mobile phone data, population mapping, remote sensing, satellite images, spatiotemporal analysis, urban air mobility, urban analytics, very high resolution (VHR) imagery.

I. INTRODUCTION

IN RECENT years, the need for precise dynamic population mapping solutions has become more and more important to convey with the changing patterns of spatial dynamics. Rapid urbanization has intensified the complexity of city management, creating a critical need for detailed knowledge of human activity dynamics to support public policy, transport, health, and urban planning. In this context, the emergence of spatio-temporal Big Data and advanced information technologies has opened new possibilities for analyzing population dynamics [1] and detecting urban conflicts [2]. Over the last decade, the exploitation

of geographically located Big Data has enabled the continuous collection of activity and mobility information with high spatio-temporal resolution, facilitating longitudinal studies that monitor both short- and long-term changes in citizen behavior [3].

This drive for precise population metrics sits within the broader scientific trend of leveraging deep learning and remote sensing to decode the flow of smart cities. These technologies have already transformed our understanding of the physical and environmental layers of urban systems. For instance, advanced inference models utilizing visual and dynamic graphical neural networks are now employed to predict vehicle behavior in high-speed scenes, significantly enhancing traffic safety and efficiency [4]. Similarly, the integration of satellite data has proven critical for sustainable development, enabling the continuous assessment of urban surface thermal environments on a global scale [5], [6] and identifying the drivers of land surface temperatures across different urban functional zones [7].

However, applying this level of precision to the human component (specifically, dynamic population mapping) remains a challenge. While foundational work has demonstrated the power of mobile network data (MND) for mapping population presence at national or regional scales [1], shifting to granular, neighborhood-level analysis requires overcoming the inherent resolution limits of single-source data [8]. To address this, recent research has increasingly turned to data fusion, combining the broad coverage of crowdsourced mobility data with the spatial precision of remote sensing to bridge the gap between coverage and granularity [9]. Just as Big Data is used to monitor the environmental and vehicular flow of the city, there is a critical need to apply similar high-resolution multisource techniques to distinguish where people are actually located within the urban area.

Traditional population mapping often relies on data from surveys (e.g., census, travel surveys, etc.), which lack temporal and behavioral information. Using census data implies the unrealistic assumption that people spend 24 h in their residence area, ignoring the dynamics of daytime activity, which can lead to significant biases in the estimation of exposure.

Multiple approaches have been discussed to analyze dynamic population presence and mobility by leveraging geographically referenced remote sensing devices and data sources. A number of studies highlighted the potential of satellite imagery, MND, GPS traces from vehicles, mobile apps and wearable devices, Wi-Fi and Bluetooth sensors, and other IoT and smart city infrastructure data.

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Satellite imagery and aerial images have been used for population mapping [10], [11] or urban planning [12]. The images require both to be available anywhere and to have enough resolution to distinguish pedestrians on the ground. Commercial earth observation satellite applications (e.g., Maxar, Pléiades, and Sentinel) provide the required resolution with a daily revisit of the location of interest. Aerial images can often provide higher resolution given the shorter distance to the ground. Nowadays, drones have gained considerable interest as a remote sensing platform for acquiring and processing images due to their cost efficiency and availability in any situation, such as emergency operations [13].

Mobile phone network data [Call Detail Records (CDRs), location-based services] has been used for a large variety of use cases related to population mapping: for road traffic [14] and mobility monitoring [15], [16]; for urban planning [17], [18]; or emergency management [19], among other examples. Data passively collected from mobile networks have proven to be highly effective in accurately capturing population presence and mobility patterns. As a result, they have been widely adopted by both public administrations and private companies for conducting mobility studies. It provides continuous information from a large sample of the total population with virtually no biases and at a much lower cost than traditional methods. The main limitation of this technology comes from the accuracy of the location: depending on the mobile network antenna's coverage area, it may range from 200 m to several kilometers.

GPS traces from sources such as vehicles, mobile apps, and wearable devices are used for improving public transport [20]; discover activity patterns [21]; studies related to the mobility of certain population segments [22]. Geolocation information from mobile apps is extremely precise, since it corresponds to GPS data [23]. This information offers small sample sets and is time-discontinuous. GPS tracking consumes significant battery, leading apps to use it only when active. Consequently, users are detected with intermittent traces. Within mobile apps, social media data is an interesting data source for understanding special events population [24], facility popularity assessment [25], or evaluation of urban green areas [26], given its combination of geolocated data and unstructured data (text, photos, or videos), providing further insights into human behavior.

Other methods rely on Wi-Fi, Bluetooth, and other IoT sensors in urban environments, which are used for estimating presence at mass events [27], [28], emergency management [29], smart mobility [30], [31], and monitoring of population social impacts such as noise [32]. Other smart city infrastructure data (traffic cameras, connected vehicles, public transport smart cards) is used for determining mobility behavior shifts in public transport [33] or improving safety on the road [34], [35]. Despite the benefits of these insightful data sources, the penetration in all segments of the population is low [36], and the deployment is uneven across different regions at this moment [37].

Understanding where and how people move or gather, whether inside buildings or in open environments, adds depth and context to simple location-based population estimates. Distinguishing between indoor and outdoor activities allows for real-time and situation-specific population density estimates,

improving accuracy. This information could be key for experts in urban planning, emergency management, public health crises like pandemics, and other aspects of smart cities, which rely on real-time data to manage traffic, public transport, and services. New mobility means a call for highly precise data to plan their operations. For example, in the current context of increasing Innovative Air Mobility operations, dynamic data of pedestrians provides key insights to plan flights in order to reduce ground risk and other impacts of this new mobility, such as noise, visual pollution, and privacy concerns.

However, current dynamic population monitoring methods struggle to effectively differentiate between indoor and outdoor environments at a large scale due to inherent trade-offs in spatio-temporal resolution and data accessibility. High-precision sources like GPS traces or IoT sensors often lack the continuous, representative sample needed to generalize outdoor activity patterns across a city, while standard mobility metrics fail to resolve the fine spatial granularity required to separate pedestrians from building occupants. In this context, integrating the extensive coverage and high temporal resolution of MND, which captures the movement of the general population, with spatially precise data overcomes the limitations of single-source methods. This fusion facilitates the accurate, large-scale monitoring of outdoor activities essential for applications such as smart city planning.

To overcome these constraints, this article proposes a methodology that strategically fuses complementary data sources to enhance spatial and temporal accuracy. In particular, this article proposes a methodology to:

- 1) **Refine user location during activities extracted from MND by combining it with GPS records from mobile apps.** This approach leverages the strengths of each data source: large datasets provide continuous longitudinal information, while smaller sample sets offer detailed spatial-temporal insights for specific times of day.
- 2) **Identify outdoor activities in urban environments.**
- 3) **Validate our findings with pedestrian data detected from satellite images.** Different methods could be used to validate the population mapping approach proposed, such as pedestrian counts available from footfall sensors (usually available as open data for many European cities) or population mapping through very high resolution (VHR) satellite imagery. The first provides a continuous source of validation, but only for the people who pass through a landmark, making it difficult to compare with a map-like approach. In this study, we have considered the validation with satellite imagery since the comparison does not require strong assumptions on the flow of pedestrians through a specific street.

II. RELATED WORK

There are several methodologies for inferring activities from MND [38]. Most studies relying on MND to analyze population dynamics estimate user positions at the Voronoi polygon level and subsequently assign them to zones of a predefined zoning system (e.g., census tracts, transport analysis zones, regular

grids) that intersect each polygon. In some cases, the spatial accuracy provided by the Voronoi level is sufficient for the study's objectives. For instance, Deville [1] compares dynamic population densities derived from CDRs in Portugal and France with those obtained from census data through a dasymetric model that incorporates ancillary sources such as land use, OpenStreetMap infrastructure, satellite nightlights, and slope. The authors conclude that even when Voronoi polygons are taken as the minimum spatial granularity, CDR-based estimates outperform those from dasymetric census models in terms of accuracy.

When higher spatial precision than the Voronoi level is required, or when specific zoning systems are needed, alternative methods are applied. These may be based on distance, land use, or density criteria [39], [40], [41], [42]. However, land use data, while widely employed, often lose discriminatory power when analyzing small urban areas, such as dense city centers, where land uses are largely homogeneous.

In [43], the authors propose a methodology to infer origin–destination (OD) trips, including trip purpose and time-of-day, from anonymized mobile phone data. This approach offers a scalable and cost-effective alternative to traditional travel surveys, showing the potential of mobile data to capture large-scale dynamic mobility patterns. Its main limitations are tied to the spatial and temporal resolution of phone records, which may miss short trips or misclassify purposes, and to the reliance on ground-truth surveys for validation, which limits generalizability.

More precise localization can be achieved through signal triangulation, which uses the position of three or more antennas and the signal delay to each, in a process analogous to GPS/GNSS positioning. In [44], various methods are compared, with Enhanced Triangulation (ETLA) offering the best balance between accuracy and cost (≈ 100 m average error), outperforming fingerprinting (149–191 m) and hybrid techniques (125–136 m). Limitations remain, particularly channel variability and the need for smoothing and static detection. Fingerprinting, though robust in theory, is constrained by the high costs of drive testing, device sensitivity, and the significant computational effort required to model complex urban environments.

The integration of MND with mobile app location data has also been explored. In [45], a methodology is presented to enhance activity–mobility diaries derived from mobile phone data with app-based location information, enabling detailed reconstruction of passenger itineraries within Palma de Mallorca Airport (Spain). A precursor to this work is [46], where GPS traces from mobile apps were used to refine the spatial accuracy of activities initially identified via MND. The present study builds upon this line of research, further analyzing and filtering app usage data to obtain a more accurate and realistic population distribution throughout the day, distinguishing between indoor and outdoor activities, and validating the results against pedestrians detected in VHR Earth observation imagery.

Population mapping with VHR satellite imagery has been addressed in several studies. In [47], the authors propose a methodology that integrates multisensor remote sensing images (e.g., Landsat, nighttime lights) with point-of-interest (POI) data

to generate high-resolution maps. By combining land cover classification, human activity proxies, and spatial modeling, this approach distributes population more precisely across urban and rural areas, outperforming census-based and single-source methods.

Recent advances in convolutional neural networks (CNNs) have also boosted automated pedestrian detection in satellite imagery [48], [49]. Nonetheless, challenges remain in reliably identifying pedestrians within complex urban environments, especially when the ultimate aim is to assess visual pollution caused by drones, as examined in [50].

Other works have combined MND with satellite-derived information. In [51], Sentinel imagery [52] is used to identify settlements, which then guide the redistribution of MND-detected population within antenna coverage areas, analogous to the way land use data are applied in earlier studies.

To the best of our knowledge, this study represents the first attempt to combine MND with mobile app GPS data and to validate the resulting models against CNN-based pedestrian detection from VHR satellite images.

III. DATA AND STUDY AREA

This study relies on three complementary datasets from a geographical perspective.

The first dataset is the MND database, accessible through long-term commercial agreements with various European mobile network operators (MNOs), including Orange Spain, Orange Belgium, and Vodafone U.K. This serves as the primary source for identifying population activities.

To enhance this perspective, we include a second dataset from Pickwell, a company that collects mobile application usage data and device-level location information from its subscribed users. Notably, the MND dataset offers broad, large-scale sampling across the territory, while the Pickwell dataset, though smaller, provides more detailed insights into user geographical distribution, albeit with intermittent coverage.

The third dataset comprises satellite images from two providers, Pléiades Neo and Maxar. These images help determine outdoor pedestrian locations and serve as a comparison and validation dataset for the fusion of the first two datasets.

A. Mobile Network Data

These data consist of a set of anonymized mobile phone records generated by the users in Madrid (Spain) on 28th February 2023, 31st July 2023, and 2nd June 2024, to coincide with the Earth observation images acquired. This data was obtained through a collaboration agreement with one of the three main MNOs in Spain, with a market share of approximately a 30%.

Among the MND strengths are:

- 1) The constant data generation along a day or sequence of days allows to reconstruct the mobility patterns of the population.
- 2) The homogeneous penetration of the MNO in virtually all socioeconomic groups of the population, together with the size of the sample set, grants a good representativeness of the population.

TABLE I
CATEGORIES OF MOBILE APPS USAGE DATA

Categories	Apps	Users
News & Magazines	11%	18%
Social	5%	13%
Entertainment/Games	10%	11%
Tools	9%	9%
Weather	8%	7%
Communications	5%	6%
Maps & Navigation	7%	5%
Sports	5%	4%
Finance	3%	4%
Dating	2%	4%
Video Players & Editors	2%	4%
Productivity	7%	3%
Lifestyle	6%	3%
Business	4%	2%
Health & Fitness	4%	2%
Beauty	3%	2%
Education	4%	1%
Travel & Local	2%	1%
Art & Design	1%	1%

Users can actively generate CDRs by interacting with the mobile network, such as making or receiving calls, sending messages, or initiating Internet connections. Alternatively, records can be generated passively through events captured by network probes. Each record includes an anonymized user identifier, a timestamp, and the identifier of the connected cell tower.

The MNO also provides the geographical locations of these towers. This data allows for the reconstruction of users' approximate locations at specific times. However, it is important to note that the records do not provide exact user positions, as each device may be located anywhere within the coverage area of the serving cell.

B. App Usage Data

GPS data obtained from mobile apps has a high spatial resolution and provides detailed information about the location of the devices. This data is aggregated from different applications across several categories. The dataset provided by Pickwell includes registers from more than 500 apps, which are distributed within the categories in Table I.

Among the information provided, the relevant fields for the methodology are: mobile device identifier, timestamp, and longitude and latitude coordinates.

Given its low sample size (less than 2% of the population), we need large periods of data in order to get a representative picture of the population distribution. Then, several months of GPS data in the municipality of Madrid are needed to properly model the people's geographical distribution [46]. GPS registers for the Madrid municipality in the following months were provided:

February 2023, July 2023, August 2023, October 2023, January 2024, April 2024, May 2024, June 2024, and October 2024.

C. Earth Observation Images

VHR satellite images capture detailed views of the Earth's surface with spatial resolutions below 1 m.

The Pléiades Neo constellation [53] offers 15 cm panchromatic images with daily global revisit capability and optimized tasking for rapid response and time-sensitive acquisitions. Images are available around 11 am local time, with a swath width of 14 km at nadir (extendable via mosaicking), suitable for large-area mapping and targeted acquisitions. Pléiades Neo images (15 cm resolution) were captured over central Madrid on February 28 and July 31, 2023, to contrast winter and summer conditions in terms of sun angle, shadow length, and occlusion from buildings.

We also acquired an image from the Maxar constellation [54], which has the largest commercial archive of VHR imagery and similar specifications to Pléiades Neo. The Maxar image was taken on June 2, 2024, also at 15 cm spatial resolution. This different provider was selected to test the transferability of the method.

D. Study Area

The MND and mobile app usage data were retrieved for the municipality of Madrid, Spain. This region has a surface area of 604 km². Due to its vast extension, we considered for the study a 9.57 km² rectangle covering the city center area, in which 105 Voronoi polygons are defined, representing the mobile cell areas in the MNO network (see Fig. 1).

The Earth observation satellite images cover an area of 27 km², due to a minimum purchase area of 25 km². The polygon was user-defined to cover different typologies of the city.

The validation between both metrics was performed in specific urban squares of the city (see Fig. 2). The squares were selected after analyzing the visibility of the satellite images due to building shadowing.

IV. METHODOLOGY

The following diagram (Fig. 3. Methodology overview.) illustrates the *integrated methodology* developed for deriving and validating outdoor people detection. This process is structured around three main data streams: *Nommon's Population Insights*, *GPS Data Preprocessing*, and the *EO Images Module*. The subsequent sections will detail each module and its specific role in the overall framework.

A. Dynamic Population Mapping

1) *MND Processing*: To analyze the presence of population in different areas of the city at different moments in time, we relied on Nommon's Population Insights tool, a state-of-the-art software platform that processes MND and blends them with

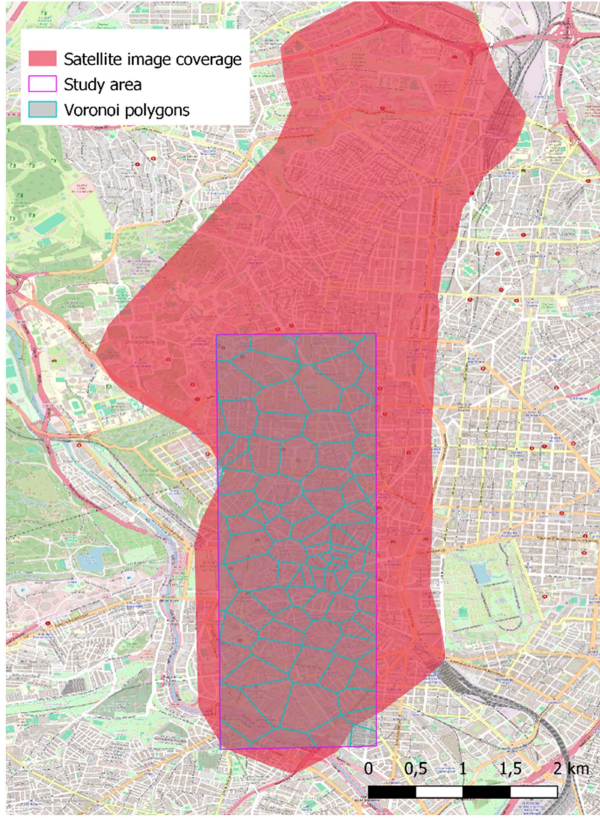


Fig. 1. Satellite image coverage area, study area, and Voronoi polygons representing the mobile network antennas coverage areas inside the area of study.

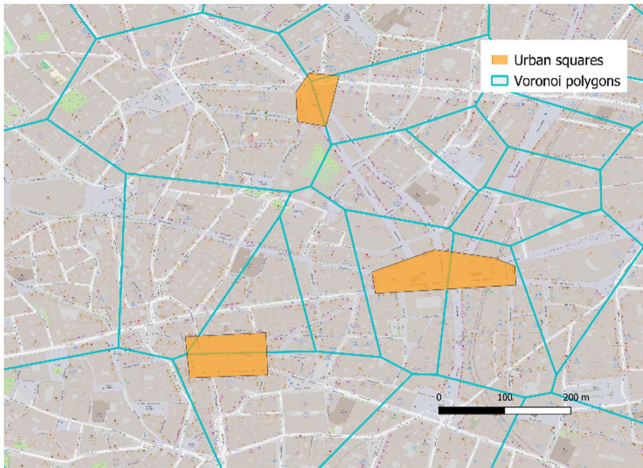


Fig. 2. Urban squares used for validation (Plaza Mayor, Plaza de Callao, Sol) and Voronoi polygons covering them.

other layers of information (e.g., land use and points of interest, sociodemographic statistics, people and traffic counts) to provide dynamic population maps, floating population statistics, and other customizable indicators. Population Insights has been used for a wide variety of applications, including the assessment of population exposure to air pollutants [55] and the use of public spaces [41].

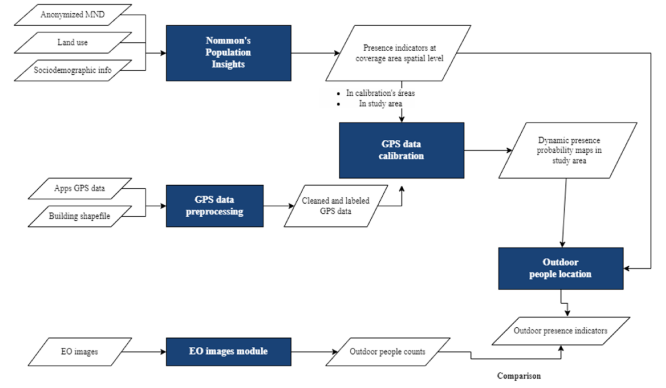


Fig. 3. Methodology overview.

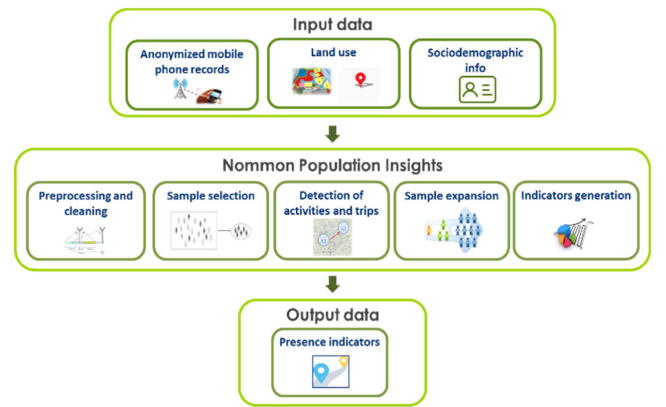


Fig. 4. Population Insights tool diagram.

The data processing pipeline involves the following five main steps, as depicted in Fig. 4.

- i) **Data pre-processing and cleansing:** formatting, elimination of useless data, error detection and fixing.
- ii) **Sample selection,** to filter users from whom it is not possible to extract reliable information (e.g., because the mobile device is switched OFF for a significant part of the day).
- iii) **Reconstruction of the sequence of activities and trips** performed by each user throughout the day. We define an “activity” as an interaction or set of interactions with the environment that occurs at a specific location and serves as the motivation for an individual’s presence there. A “trip” is understood as a sequence of one or more journeys (also referred to as “stages” or “legs”) connecting two consecutive activities. In this framework, a trip is characterized by a main purpose, typically associated with the activity at its origin and/or destination. To identify activities and distinguish them from intermediate stops within a single trip, we apply a set of criteria to multiday mobility patterns. These criteria include the frequency of visits, time of day, and duration of stay at observed locations. This analytical approach enables the separation of meaningful activities from transient pauses, offering a clearer understanding of individual mobility behavior. Each activity is described by its location, start and end times, and type, classified into

categories such as home, work, study, other frequent activities, and nonfrequent activities [56]. This classification relies on the analysis of longitudinal behavioral patterns over several weeks or months. For instance, a user's place of residence is inferred as the location where they most frequently spend the night. Once an activity is identified at the mobile antenna level, at this point there are two options to refine the user's position within the antenna's coverage area, approximated using Voronoi tessellation: using a probabilistic approach based on land use or a more accurate probabilistic positioning based on high-resolution geospatial data, as we present in this article. Each trip is described by its origin and destination, corresponding to the locations of the preceding and subsequent activities, as well as the start and end times. Additional attributes include the location and timing of intermediate stops (if any), along with the mode of transport and route chosen for each trip stage.

- iii) **Sample expansion** through the use of reference sociodemographic statistics (e.g., census data) and other reference data (e.g., people and traffic counts). The expansion factor is computed at the district level as the ratio between the total number of residents in the district, based on official census data, and the number of users in the sample assigned to that district. This approach enables the adjustment for any potential spatial heterogeneity in the MNOs market share, ensuring that population estimates more accurately reflect the true distribution of residents across districts.
- v) **Generation of presence indicators** with the required spatial, temporal, and sociodemographic granularity, ensuring that the level of aggregation guarantees compliance with GDPR by preventing any risk of reidentification of individuals.

However, this data presents the limitation of low spatial resolution. Location is subject to the coverage areas of the mobile network antennas, which vary from some hundreds of meters to kilometers.

In mobile network planning, Voronoi diagrams are employed to represent the coverage areas of cell towers, with each cell depicted as a Voronoi polygon. These polygons involve the regions that are geographically closest to a particular tower relative to the rest of the network. Fig. 5 shows the Voronoi polygons that define antenna coverage areas.

This low spatial granularity, which reaches some hundreds of meters in the city center, where the resolution is best, falls short for some location analysis that aims to disaggregate between indoors and outdoors.

In densely populated areas, mobile antennas are close to each other, in order to avoid shadow areas and provide proper coverage to the entire population connected. The representation of coverage areas with Voronoi polygons in this case may fall short, since other factors as orientation, play a role, and there are overlaps between coverage areas. In addition, it may happen that the study area lies on the intersection between Voronoi polygons. To consider MND more robust, the spatial aggregation of Voronoi polygons provides more reliable values on the number of people present in a region.

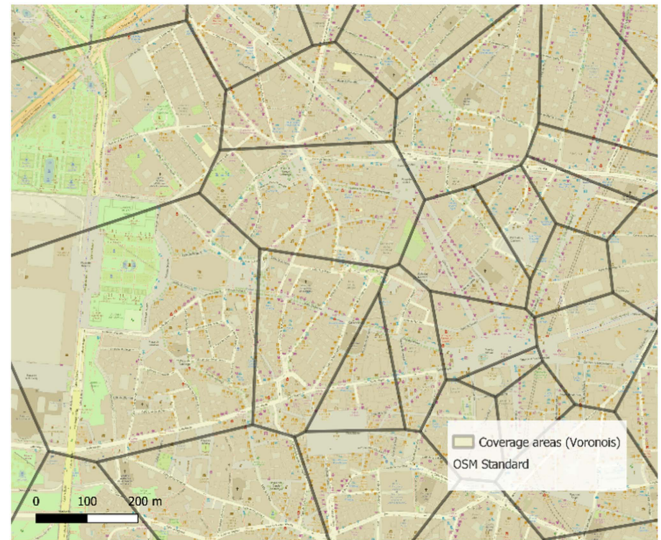


Fig. 5. Coverage areas of mobile phone antennas approximated by a Voronoi tessellation in the Madrid center.

The procedure is as follows:

- 1) *Identify relevant Voronoi cells*: We first selected the Voronoi cells that strictly cover the area of interest.
 - 2) *Evaluate population distribution*: For each selected Voronoi cell, we analyzed MND population indicators in its adjacent Voronoi neighbors. Large gradients in population between adjacent cells are further explored. We compare the MND population with GPS data and expert judgement (e.g., based on land use, buildings, public spaces) to assess whether such population concentration made contextual sense.
 - 3) If the high presence could not be justified spatially, we merged the Voronoi cell with one or more of its adjacent neighbors to form a larger, more representative coverage unit.
 - 4) This procedure was repeated iteratively for the remaining Voronoi cells covering the area.
 - 5) We avoid very large clusters. The wider the cluster, the more spatial homogeneity we assume in the population presence.
- 2) *App Usage Data Processing*: As mentioned before, GPS mobile app usage data was purchased in raw format, and therefore, the individual traces can present temporal discontinuities. The processing algorithm is recorded in Algorithm 1. The following five key preprocessing stages have been identified to ensure the quality of the analysis:
- a) **Discard GPS records where either the latitude or longitude is rounded to two decimal places.**
 - b) **Removal of simultaneous records** generated by the same user. This is just a sanity check, and typically discard just a few records.
 - c) **Removing unusually high-density coordinates.** This preprocessing step identifies and removes GPS coordinates that are likely to be artificial or nonhuman locations, such as map defaults, cell tower centroids, or anonymization artifacts. In other words, we detect high-density coordinates, flag them as “nonhuman” locations, and exclude

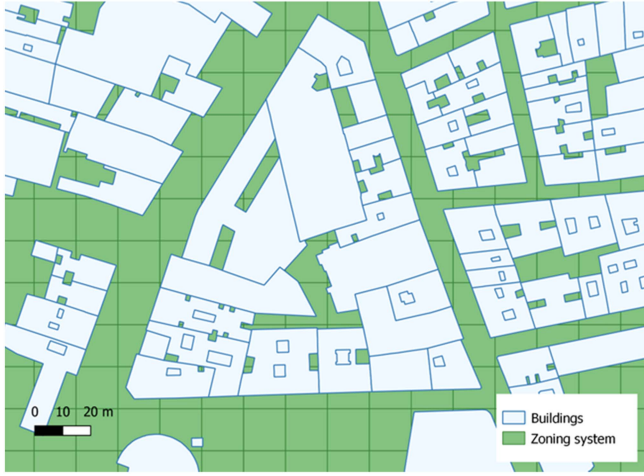


Fig. 6. Building shapes and regular mesh covering the outside areas.

them from the dataset. In order to remove abnormally high-density coordinates, we discard points that share exactly the same location (with all the decimals) as more than five other GPS points.

- d) **Filtering of trajectories incompatible with pedestrian movement**, by discarding sequences of GPS records whose instantaneous speed exceeds 20 km/h.
- e) **Detection and classification of indoor and outdoor chains**. In some cases, sequences of GPS registers exhibit spatial uncertainty between indoors/outdoors. A longitudinal analysis of user trajectories can help infer whether a user is realistically located indoors or outdoors. In this analysis, we chunk the user trajectory into chains. We define a chain as the sequence of registers in which the time difference between consecutive registers does not exceed a threshold. Additionally, we break a chain if the user goes from indoors to outdoors (based on building shape data depicted in Fig. 6) and the time spent in the transition is enough (3 min). The concept of chains is a tool that allows for avoiding jumps in the signal. Fig. 7 illustrates an example trajectory where the GPS signal briefly places a user inside a building while they are actually walking down the street. Despite this outlier, the surrounding registers and timing allow us to treat the movement as part of a single outdoor chain.

3) *Data Fusion*: We expose in this article a new version of Population Insights developed in the context of the SESAR3 ER project MUSE¹. The methodology takes advantage of the strengths found in each data source and integrates MND with GPS data to better differentiate between indoor and outdoor activities.

The differentiation between indoor and outdoor activities is performed considering the building's vectorized shape provided by the city council of Madrid. We have focused on outdoor precise positioning, so the area that falls outside the buildings



Fig. 7. GPS registers in a street and its surroundings. Example of jumps in a user's signal between outdoor and indoor.



Fig. 8. Procedure to distribute the MND population detected using GPS data.

is our area of interest, divided into a square grid of 15 m side, defining our zoning system.

The goal is to create a dynamic probability map in the zoning system. For each coverage area, and for each time interval, we apply the Laplace rule to estimate the instantaneous probability of being somewhere in the zoning system (see Fig. 8). The calculation is as follows: suppose we want to calculate the probability of being in the zone of interest called z , which overlaps with the coverage area $ca(z)$ at hour h . The total set of all available GPS registers is denoted G , and each register g has a location, g_{loc} , and a timestamp, g_t . Let S_z be the set of GPS registers located within the zone of interest z during the time interval $[h-2, h+2]$

$$S_z = \{g \in G \mid g_{loc} \in z \text{ and } h-2 \leq g_t \leq h+2\}. \quad (1)$$

Let $S_{ca(z)}$ be the set of GPS registers located within the larger coverage area $ca(z)$ during the same time interval

$$S_{ca(z)} = \{g \in G \mid g_{loc} \in ca(z) \text{ and } h-2 \leq g_t \leq h+2\}. \quad (2)$$

Then, the probability of $P(z, h)$ can be expressed as the ratio of the number of elements in these two sets

$$P(z, h) = S_z / S_{ca(z)}. \quad (3)$$

The time interval is typically an hour. However, in order to make the probability maps robust to noise at specific hours, we consider time windows centered at the hour of interest.

¹ <https://musesesarproject.eu/>

The dynamic nature of the presence probability maps is essential to reflect the behavioral patterns (day of the week, seasonal effect, holidays...) of the MND indicators. In the experiments that are described in this article, we have aggregated GPS following the similarity criteria of a typical weekday, not affected by a nearby holiday, choosing among the available GPS dates. The complete process to assign locations to the population detected with MND is described in Algorithm 2.

B. Calibration

To ensure the accuracy of the spatial disaggregation of presence indicators onto the mentioned grid, a calibration process was introduced into the algorithm.

Although the integration of GPS data enables the generation of high-resolution probability maps of presence, we observed systematic biases, particularly during nighttime hours. Specifically, anomalously high presence probabilities were detected at night, likely attributable to biases in GPS data collected via mobile applications. To correct this effect and produce a more realistic spatial distribution of population presence, both temporal and spatial calibration mechanisms were implemented to adjust the observed probabilities.

The adjustment of the dynamic probability maps was carried out through comparison with a ground truth proxy derived from the mobile network indicators themselves. During nighttime hours (00:00–01:00), we used presence indicators associated with the activity type “HOME” as a reference, restricting the analysis to purely residential areas where residents have very limited options for engaging in activities other than staying indoors. Under these conditions, the assumption that the population with “HOME” activity is indoors is close to being accurate. An activity is labeled as “HOME” if it takes place in the location where it is most frequent to have periods of low activity (fewer calls and messages) corresponding to sleeping time.

For each residential area, we compute the adjustment factor f of a given area k as follows:

$$f_k = \frac{1 - p_k^{home}/p_k}{g_k^{out}/g_k} \quad (4)$$

being g_k the total GPS users detected, g_k^{out} the GPS users outdoors, p_k the total population detected with MND in the area and p_k^{home} the population detected with activity home.

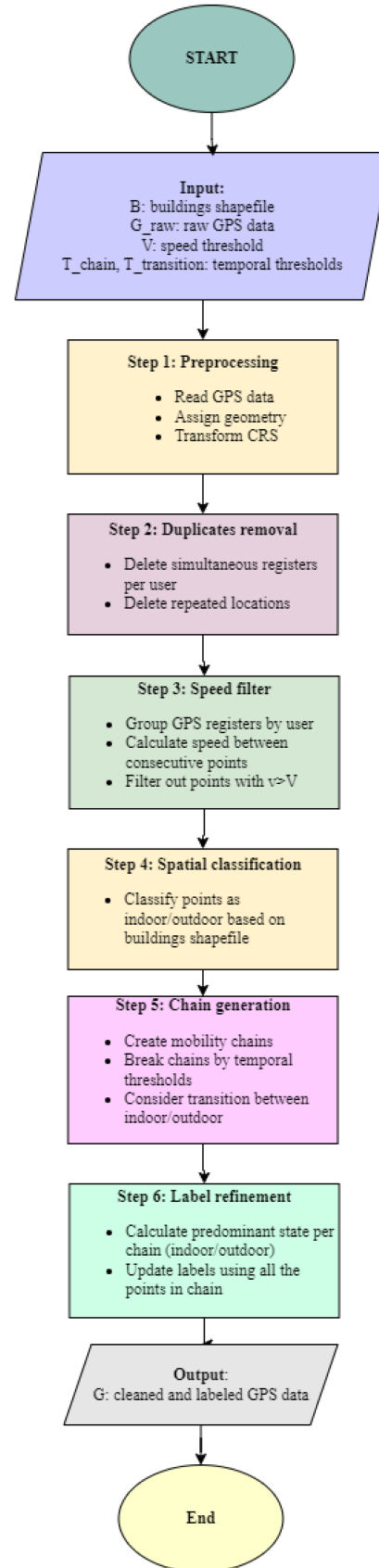
If the correction factor is stable across the different residential areas, we take the mean value of all of them.

Similarly, between 09:00 and 12:00, the same approach was applied in exclusively office areas, with no residential or leisure spaces present. In this case, the ground truth was based on indicators associated with the “WORK” activity. A user’s activity is labeled as “WORK” if it is located at the most frequently visited location (after home) during workdays.

For each office area, we compute the adjustment factor as follows:

$$f_k = \frac{1 - p_k^{work}/p_k}{g_k^{out}/g_k} \quad (5)$$

Algorithm 1: GPS Data Cleaning.



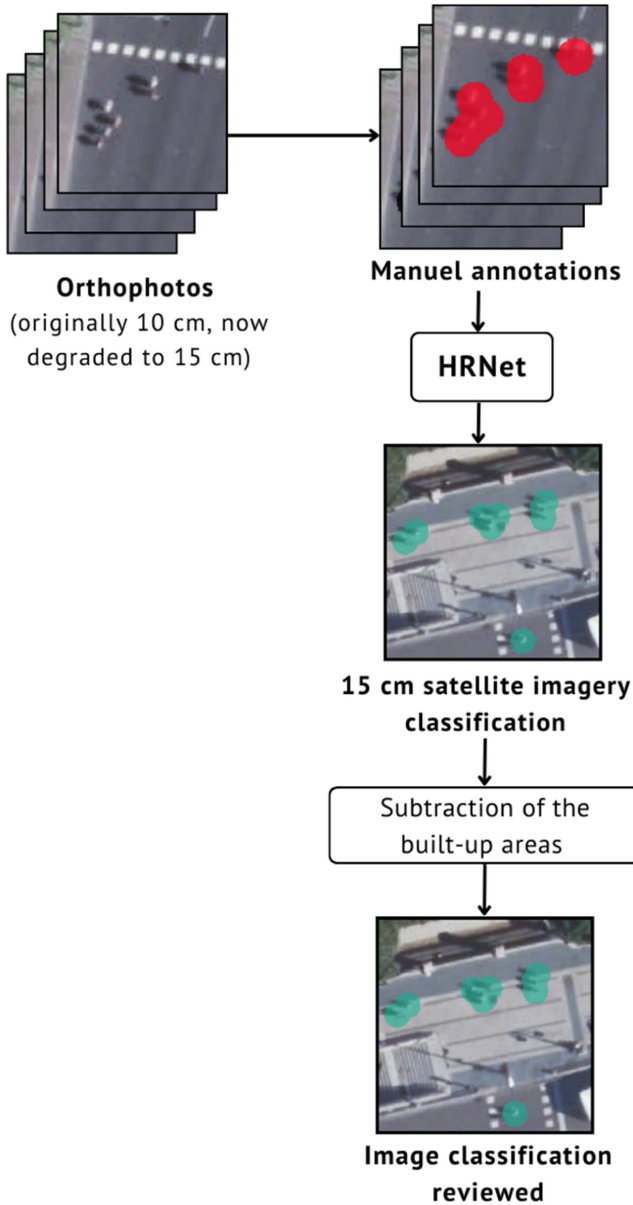


Fig. 9. Overview of the pedestrian detection from satellite images pipeline.

being p_k^{work} the population detected with MND with activity “WORK.”

This calibration strategy, based on time slots and urban typologies, allows us to scale the GPS-derived probability estimates and correct for systematic biases resulting from the dynamic and uneven nature of GPS data coverage.

C. Pedestrian Location With Satellite Images

We developed a remote sensing pipeline capable of detecting individuals from VHR satellite imagery (see Fig. 9). The workflow involves: (i) dataset construction from annotated orthophotos, (ii) model training using a High-Resolution Network (HRNet)-like architecture, and (iii) postprocessing with spatial filtering and artifact removal.

TABLE II
DATA PREPARATION STEPS

Step	Description
Input imagery	10 cm orthophotos from Madrid (downsampled to 15 cm)
Annotation method	Manual digitization of shadows (visual identification of pedestrians)
Training samples	4500+ instances over 21 images
Patch size/stride	64×64 px patches, 32 px stride
Augmentations	Flip, rotation $\pm 15^\circ$, brightness $\pm 20\%$
Data split	70% train, 15% val, 15% test

TABLE III
MODEL DETAILS

Model details	Configuration
Architecture	HRNet-inspired U-Net hybrid
Input shape	(64, 64, 3)
Output	Binary pedestrian presence map
Loss function	Dice Loss + Binary Cross-Entropy
Optimizer/LR	Adam/ $1e-4$
Epochs/Early stopping	100/patience = 10

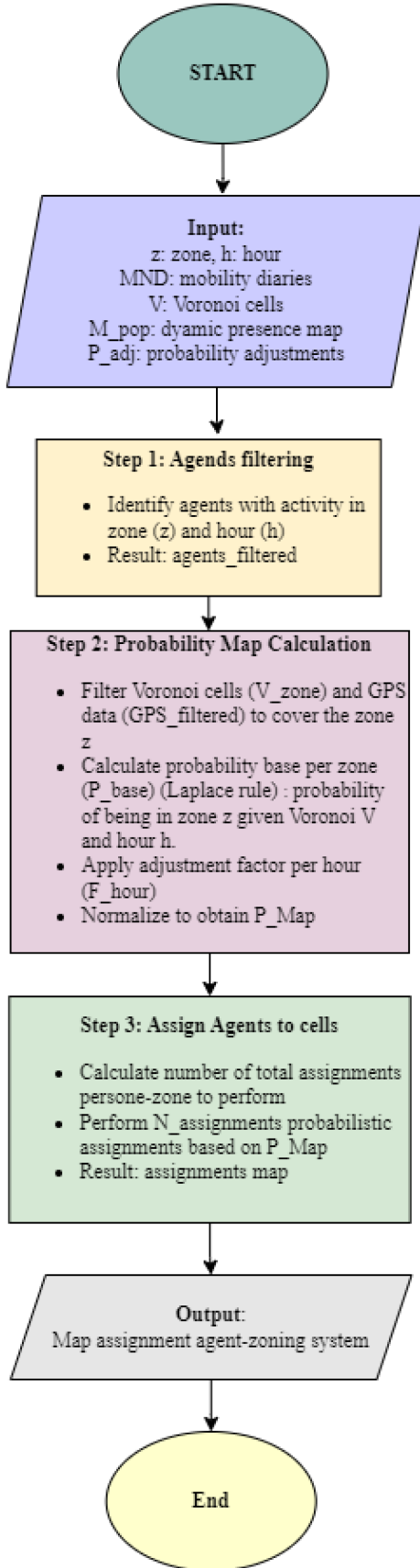
Data preparation: To address the lack of large-scale annotated training data in satellite imagery, we used freely available 10 cm orthophotos from the Madrid geoportal [57], which were downsampled to 15 cm to match the target resolution. We conducted manual annotation of 4500+ pedestrian instances across 21 orthoimages from the Madrid council geoportal, based on visual shadow identification (see Table II).

Deep learning architecture: HRNet architecture, introduced in 2019 by [58], known for preserving spatial details through multiresolution processing.

Unlike U-Net or other encoder–decoder models, HRNet maintains parallel streams at multiple resolutions, exchanging features between them at each stage. This is particularly suited to our task, where pedestrians occupy only a few pixels and are detected via their shadows. The model was implemented in Keras and trained using Dice + Binary Cross-Entropy loss, with the Adam optimizer and early stopping (100 epochs, LR = $1e-4$, WD = $1e-5$). Table III provides more details on the image detection model.

Postprocessing and spatial filtering: After initial inference, a postprocessing pipeline was applied to refine the raw predictions and reduce false detections caused by urban clutter. The steps include:

- **Land cover masking:** Model outputs were spatially constrained using the Urban Atlas dataset [59] to exclude non-walkable surfaces. Only predictions falling within classes such as pedestrian zones, parks, and sidewalks were retained.
- **Artifact removal:** Small, isolated detections likely to be noise (e.g., shadows from rooftops, lampposts, vehicles) were removed using morphological operations based on connected component size thresholds.

Algorithm 2: MND Location Assignment.

- *Centroid extraction:* The cleaned binary masks were converted into geolocated point features by computing the centroid of each connected pedestrian-shaped region, enabling integration into spatial analytics platforms.

This combination of semantic filtering and morphological cleaning significantly improved the precision of the pedestrian detection outputs, particularly in dense urban areas prone to visual ambiguity.

D. Validation Process

The validation process, based on the direct comparison between presence indicators and satellite-derived person counts, was specifically conducted in selected urban squares of Madrid. Urban squares can be considered as areas where the intrinsic limitations of both data sources are less pronounced, i.e., less area is blocked due to shadowing of the buildings nearby, they are likely to host pedestrians, and large open spaces improve MND robustness.

V. RESULTS

This section outlines the results of each phase involved in generating high spatial resolution presence indicators, as well as their validation process. The phases include: (1) preprocessing and cleaning of GPS data, (2) adjustment of dynamic probability maps, (3) preparation of presence indicators at the Voronoi polygon level, (4) pedestrian detection and counting using satellite imagery, and (5) validation results.

A. Dynamic Population Mapping

1) *Preprocessing and Cleaning of GPS App Data:* Before presenting the results of the methodology exposed in Section IV. Methodology, it is important to outline several considerations that conditioned the design of the validation exercises.

First of all, we wanted to validate the methodology at two different seasons, summer and winter. Therefore, two different sets of GPS data were needed to create two different dynamic probability maps. The curated GPS data that was ready to use in the pipeline can be seen in Figs. 10 and 11.

The dataset filtering removed the following number of records:

- Low spatial precision:* Discarded approximately 25% of the sample. The rounding of the coordinates shows a clear pattern in the data, see Fig. 12.
- Unusually high-density coordinates:* Discarded about 1.5% of the remaining sample.
- Trajectories incompatible with pedestrian movement:* Discarded around 1.2% of the remaining sample.

In total, after all filters were applied, approximately a 27% of the initial sample was discarded.

2) *Adjustment of Dynamic Probability Maps:* The spatial scope of the residential areas used for adjustment is selected within the municipality of Madrid, focusing on the following zones: Arturo Soria, Carabanchel, San Chinarro, Valdeaceras, and South of Madrid. The Voronoi polygons used for the San

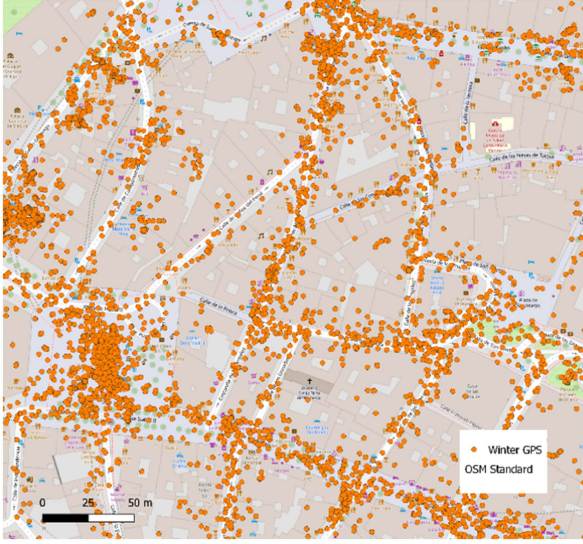


Fig. 10. GPS points recorded during a winter period to capture the specific patterns of the season.

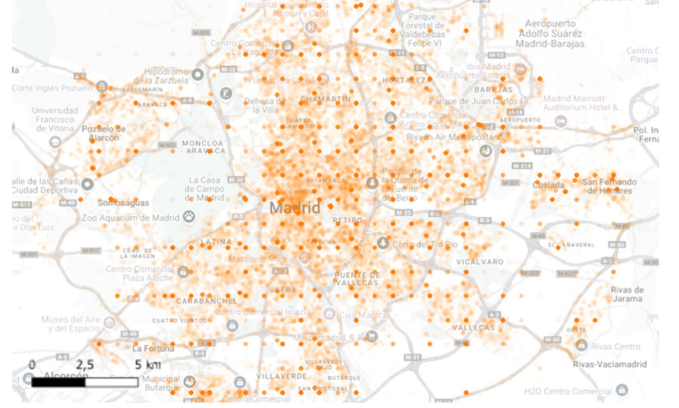


Fig. 12. Raw GPS locations in the municipality of Madrid.

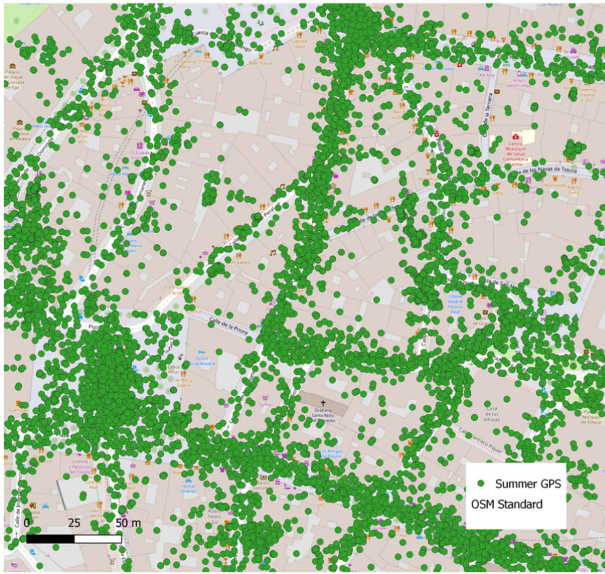


Fig. 11. GPS points recorded during a summer period to capture the specific patterns of the season.

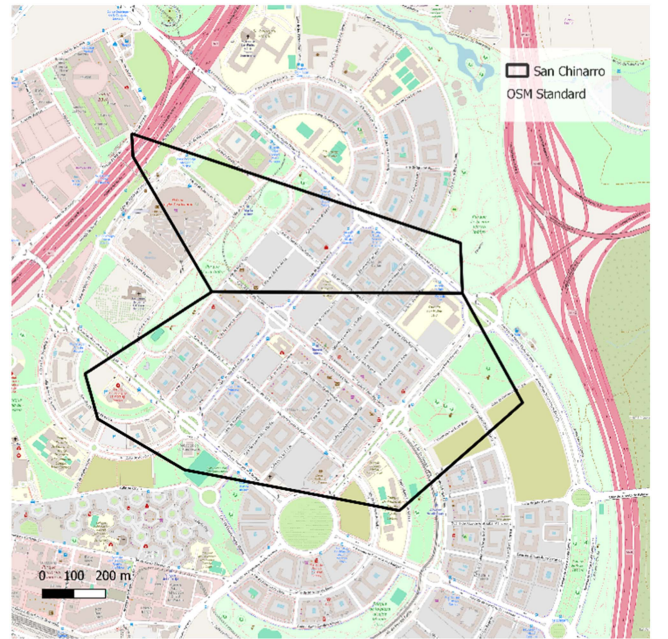


Fig. 13. San Chinarro area used for nighttime adjustment with a very residential nature.

Chinarro case can be seen in Fig. 13. Table IV shows the adjustment factors obtained for each area considered and the average factor we used to calibrate nighttime data in the rest of the city.

The areas used for daytime adjustment are focused on Ciudad de Telefónica and Ciudad BBVA, two major enterprise complexes in the city (Fig. 14). Table V shows the adjustment factors obtained for the two office areas considered and the average factor we used to calibrate daytime data in the rest of the city.

3) *Presence Indicators at Voronoi Polygon Level:* We present here the results of the experiments conducted to analyze the spatial characteristics of mobile antenna coverage, the limitations

TABLE IV
CORRECTION FACTORS OBTAINED FOR NIGHTTIME ADJUSTMENT

Area	Correction factor nighttime
Valdeaceras	0.24
Arturo Soria	0.18
Carabanchel	0.30
San Chinarro	0.21
South	0.17
Average	0.22

TABLE V
CORRECTION FACTORS OBTAINED FOR DAYTIME ADJUSTMENT

Area	Correction factor daytime
Ciudad telefónica	0.15
Ciudad BBVA	0.13
Average	0.14

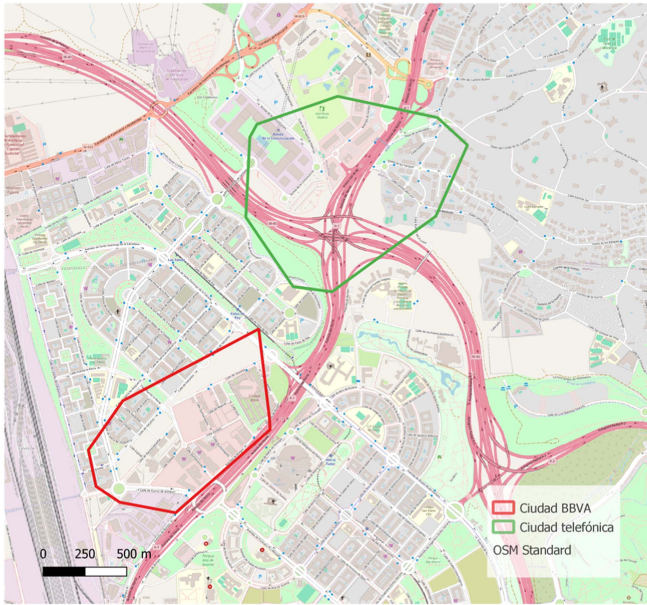


Fig. 14. Areas used for daytime adjustment.

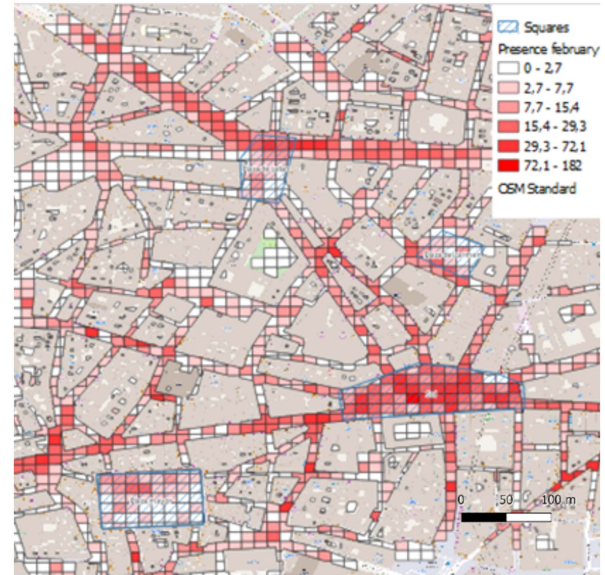


Fig. 16. Presence indicators obtained with the methodology at 11 am in February 2023.

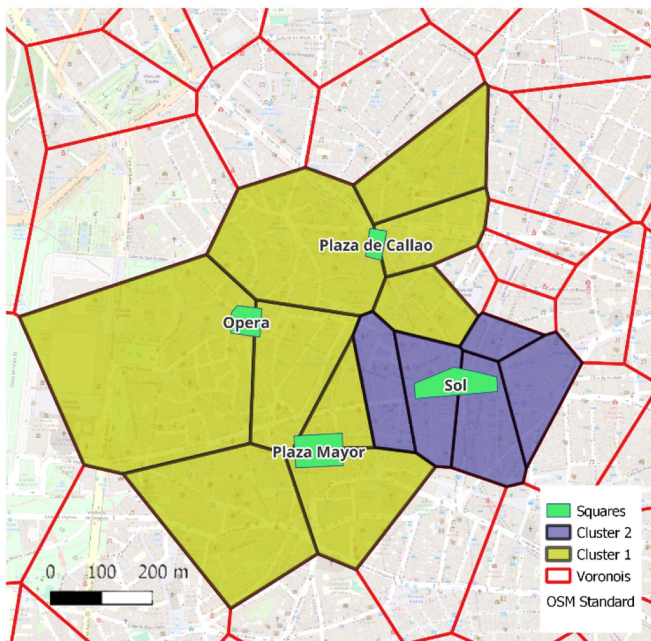


Fig. 15. Clusters of Voronoi surrounding the different squares.

imposed by urban density, and the need for aggregation strategies to improve the robustness of the MND data.

To consider this effect, we created 2 Voronoi clusters around the squares of interest, see Fig. 15.

The idea behind clustering Voronoi cells is to compensate for the presence of sinks observed in the MND indicators—areas with disproportionately high detected presence due to antenna coverage, which do not correspond to highly populated zones based on GPS data. By aggregating Voronoi cells, we aim to

TABLE VI
NUMBER OF PEDESTRIANS PRESENT OUTSIDE AT DIFFERENT TIMES

Time	Pedestrians outside
9:00	29 855
10:30	35 244
12:00	37 615
13:30	37 937
15:00	36 038
16:30	33 292

smooth out these inconsistencies and better align the MND-derived presence estimates with the actual spatial distribution of the population.

Ideally, each square would correspond to have one cluster per square to prevent certain squares—those that are much more frequently visited according to GPS data—from becoming artificial attractors in the dynamic probability maps relative to the other squares.

The previous process was manually conducted on three different dates (28 February 2023, 31 July 2023, and 2 June 2024) as the map of Voronoi evolves in time.

For each date, the fusion between the MND presence indicators at the Voronoi aggregated level and the corresponding GPS dynamic probability map (summer/winter map) resulted in a population map with a high level of spatial resolution. With this process, we obtained results for the study area presented. Fig. 16 shows the population presence outside obtained on 28th February 2023.

For the study area, the number of people present outside on the 28th of February 2023 is presented in Table VI.

TABLE VII
PERFORMANCE TABLE FOR THE THREE SATELLITE IMAGES

Image	PN Feb.	PN Jul.	Maxar
Ground truth	10941	3019	7116
Total detections	21 100	23 342	11 603
True positives (TP)	7705	2441	2627
False negatives (FN)	3236	578	4489
False positives (FP)	13 395	20 901	8976
Precision = user accuracy (%)	36.52	10.46	22.64
Recall = producer accuracy (%)	70.41	80.86	36.92
Accuracy (%)	63.48	89.54	77.36
F1-score (%)	48.01	18.46	28.07

B. Pedestrian Location With Satellite Images

The pedestrian detection model was initially trained on manually annotated orthophotos from the Madrid Geoportal, originally acquired at 10 cm resolution and downsampled to 15 cm to match the target satellite imagery. This freely available aerial imagery provided high-quality visual conditions (nadir acquisition, minimal occlusion, and sharp shadow features) ideal for building a training set without incurring the high cost of annotated satellite data.

The input satellite imagery and ground truth annotations were split into 64×64 pixel patches with a stride of 32 pixels, providing dense coverage while preserving local spatial structure. The dataset was randomly divided into training (70%), validation (15%), and test (15%) subsets. To increase model generalization and robustness to illumination and scene variability, standard image augmentations were applied during training, including random horizontal and vertical flips, $\pm 15^\circ$ rotations, and $\pm 20\%$ brightness variation.

The neural network architecture was based on a modified U-Net with an HRNet-inspired design. It combined a classical encoder-decoder pathway with a parallel high-resolution stream, preserving fine spatial details throughout the network. The parallel stream maintained the original input resolution and was later fused with the upsampled decoder output to enhance feature richness. The model was implemented in Keras, using convolutional blocks with ReLU activations and batch normalization, followed by transposed convolutions for upsampling. A final 1×1 convolutional layer with sigmoid activation produced the binary pedestrian presence map.

Model training was conducted using the Adam optimizer with an initial learning rate of $1e-4$ and weight decay of $1e-5$. The model was trained for 100 epochs, with early stopping based on the validation loss (patience = 10). The loss function combined Dice Loss with Binary Cross-Entropy to address class imbalance and improve segmentation around pedestrian shadows.

Following training on the orthophotos, the model was fine-tuned on a set of patches extracted from Pléiades Neo imagery to adapt to differences in radiometric properties and acquisition

geometry. This step improved transferability and detection performance on real satellite images, including both the February and July Pléiades Neo scenes.

To validate model performance, we conducted a manual verification process, where a human analyst systematically reviewed each predicted detection. For each scene, true positives, false positives, and missed detections were recorded by visually confirming the presence (or absence) of a pedestrian or shadow at each predicted location, and by identifying any undetected individuals. This qualitative, image-by-image assessment allowed for accurate estimation of detection accuracy, especially in cluttered or ambiguous areas.

The main evaluation metric was the number of pedestrians correctly detected by the model, relative to the total manually identified by a human validator in each scene. This approach emphasizes practical utility over pixel-wise accuracy. For the Pléiades Neo image acquired on 28 February 2023, a total of 10 941 pedestrians were manually identified. The 31 July 2023 image contained 3019 pedestrians, reflecting reduced urban activity during the summer period and possibly fewer people outdoors due to higher temperatures. In the Maxar image acquired on 2 June 2024, 7116 pedestrians were manually annotated. These counts served as ground truth references for evaluating detection performance in each case.

The detection model achieved an estimated 70.41% detection rate for the February image and 80.86% for the July image, after postprocessing. This performance difference reflects the seasonal trade-off: winter scenes provide longer and more visible shadows, which are helpful for detection, but are also more affected by occlusions caused by tall buildings and narrow streets. In contrast, the July image showed less occlusion, leading to better visibility in open areas, but shorter and softer shadows slightly reduced contrast, see Fig. 17.

False positives were primarily associated with urban clutter such as rooftops, vehicles, and street furniture. These were significantly reduced by integrating Urban Atlas land cover data and applying a morphological cleaning step during postprocessing.

The study demonstrates that shadow-based pedestrian detection using freely available orthophotos offers a cost-effective and scalable approach, maintaining compatibility with high-resolution satellite imagery despite the challenges of seasonal changes, lighting conditions, urban structure, and capture of temporal dynamics.

These results demonstrate the sensitivity of shadow-based pedestrian detection to seasonal variations and structural characteristics of the urban environment, such as occlusions from tall buildings and the distribution of open spaces. Despite the high cost and limited accessibility of VHR satellite imagery, we addressed this constraint by leveraging freely available orthophotos from the Madrid geoportal to build a high-quality training dataset. This strategy significantly reduced the need for costly annotated satellite data while maintaining compatibility with the resolution of Pléiades Neo imagery. Together, these findings validate the feasibility and cost-efficiency of using VHR imagery for mapping human presence at scale in support of urban air mobility assessments.



Fig. 17. Detected pedestrians (yellow points) on Pléiades Neo imagery at 1/122 scale. Pedestrian shadows are more easily detectable in February (top), while they are less visible in July (bottom). Image Pléiades Neo Airbus DS (2023).

However, several limitations should be acknowledged. The approach relies on indirect detection via shadows, making it sensitive to lighting conditions, seasonality, and urban occlusions. As a result, performance can vary significantly depending on sun angle, time of year, and building geometry. While training on freely available orthophotos helps overcome the high cost of satellite imagery, applying the model to satellite data introduces a domain shift that may affect generalization. Furthermore, despite contextual filtering, false positives persist in areas with dense urban clutter. Finally, the method provides only a static snapshot of pedestrian presence, limiting its ability to capture temporal dynamics without frequent acquisition.

Fig. 18 presents a snapshot of the pedestrian presence in each satellite image analyzed.

The manual evaluation confirmed the model's satisfactory performance on the Pléiades Neo scenes from February and July 2023, despite the differences in geometry and radiometry compared to the training orthophotos. These satellite images share key properties with the orthophotos (i.e., high spatial resolution, clear shadow projection, and relatively nadir acquisition angles), which likely facilitated cross-domain generalization.

To further test the model's robustness and transferability, we applied it without any fine-tuning to a third VHR satellite image acquired by Maxar on 2 June 2024, also at 15 cm resolution. In this case, the model yielded a significantly lower accuracy of 36.92%, highlighting the limitations of direct cross-sensor

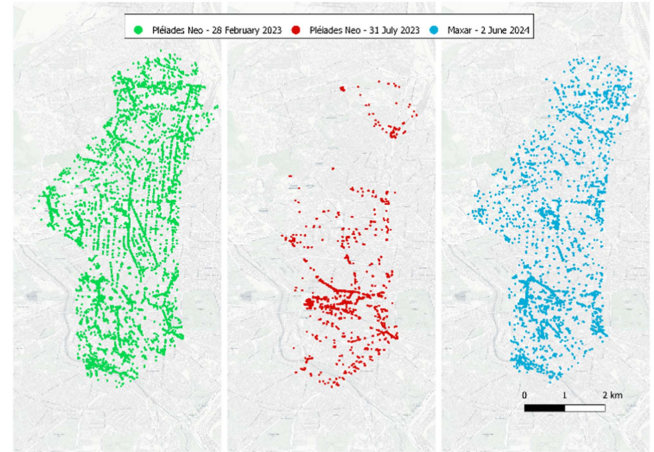


Fig. 18. Comparison between pedestrians detected in the three different orthophotos analyzed.

application. This performance gap can be attributed to the combined effects of a more pronounced domain shift (differences in spectral response, radiometric profile, and scene characteristics), as well as the different shadow lengths due to high solar elevation near the summer solstice. The full performance metrics can be found in Table VII.

The detection results presented in the maps incorporate manual validation to ensure that only visually confirmed pedestrian instances are reported.

These results emphasize the importance of domain adaptation strategies or sensor-specific fine-tuning when extending the method to other satellites or geographic contexts.

Despite similar spatial resolution, the two images in Fig. 19 exhibit notable differences in acquisition geometry. The Maxar scene, acquired with a higher off-nadir angle, presents distorted building shapes, longer but softer shadows, and reduced radiometric contrast, making pedestrian detection more challenging without fine-tuning. In contrast, the Pléiades Neo image, closer to nadir, offers more regular geometry, sharper shadow definition, and higher overall contrast.

We analyzed shadowed and visually obstructed areas in each satellite scene using a two-step supervised classification approach based on Random Forest (RF) models. The first model was trained to discriminate between shadowed and illuminated pixels, while the second targeted tree crowns, both using radiometric features from the visible bands. The outputs were combined to produce a binary occlusion mask encompassing both cast shadows and vegetation-related obstructions.

To evaluate how these occlusions impacted pedestrian detectability, we intersected the occlusion mask with road segments from the Urban Atlas dataset. This allowed us to compute, for each street segment, the proportion of surface affected by occlusions, either from building shadows or overhanging tree canopies. This analysis was critical to identify areas where pedestrian presence cannot be reliably detected due to a lack of direct illumination or visual obstruction. By flagging heavily occluded streets, we were able to better interpret apparent

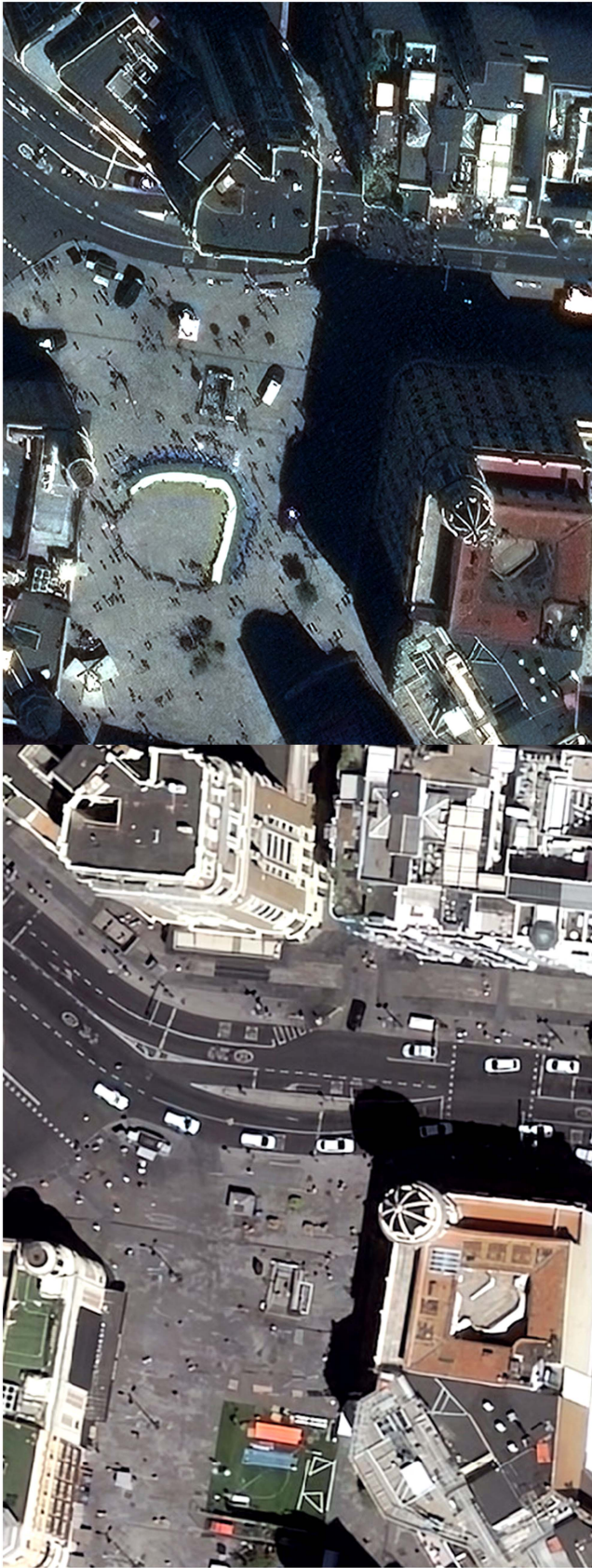


Fig. 19. Comparison between a Maxar image (June 2, 2024) on top and a Pléiades Neo image (July 7, 2023), below, over Plaza de Callao, both at 15 cm resolution.



Fig. 20. Shadow occlusion analysis over the February 2023 Pléiades Neo image.

TABLE VIII
VALIDATION RESULTS: COMPARISON BETWEEN MND+GPS WITH THE
SATELLITE IMAGES DETECTED PEDESTRIANS

Urban square	Technique	Date		
		Feb. 2023	Jul. 2023	Jun. 2024
Ópera	MND	113	129	107
	MND+GPS	54	39	37
	Satellite	46	23	43
Plaza de Callao	MND	90	139	115
	MND+GPS	72	53	47
	Satellite	70	49	133
Plaza Mayor	MND	362	270	225
	MND+GPS	266	49	51
	Satellite	221	83	128
Sol	MND	339	279	228
	MND+GPS	392	239	199
	Satellite	201	140	161

absences, adjust detection expectations, and refine our overall assessment of observable pedestrian presence in urban environments. In Fig. 20, the overlaid vector layer shows zones classified as shadowed or structurally occluded, intersected with the Urban Atlas road network to highlight street segments where pedestrian detection is not feasible due to a lack of direct illumination.

C. Validation Results

We selected for the validation some urban squares placed in Madrid: Ópera, Plaza de Callao, Plaza Mayor, and Plaza del Sol. These squares fall inside the area of the satellite images obtained and have a high number of pedestrians at every moment of the day, since they are placed in the city center, in very touristic locations. Also, the squares were characterized as areas with high visibility, i.e., the shadows of the buildings do not cover more than a 20% of the area of the square.

Table VIII and Fig. 21 summarize the results obtained employing both methodologies. We can observe a reasonable level of agreement between MND+GPS and the satellite-detected pedestrians, supporting the idea of using MND enhanced with

TABLE IX
STATISTICAL COMPARISON BETWEEN MND+GPS AND MND WITH THE
SATELLITE IMAGES DETECTED PEDESTRIANS

Urban square	Comparison	Pearson	RMSE
Ópera	MND vs Satellite	-0.92	81.28
Ópera	MND+GPS vs Satellite	0.51	10.89
Plaza de Callao	MND vs Satellite	-0.22	54.32
Plaza de Callao	MND+GPS vs Satellite	-0.40	49.71
Plaza Mayor	MND vs Satellite	0.79	146.35
Plaza Mayor	MND+GPS vs Satellite	0.95	55.10
Sol	MND vs Satellite	0.68	119.51
Sol	MND+GPS vs Satellite	0.85	126.12

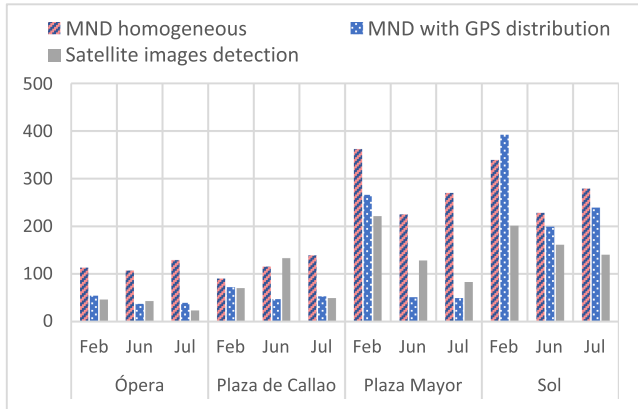


Fig. 21. Validation results: comparison between MND+GPS with the satellite images detected pedestrians.

GPS for continuous monitoring of the population inside and outside.

The integration of GPS data (MND+GPS) proves to be clearly superior to MND alone in terms of absolute error (MAE, RMSE) and relative error (MAPE). In absolute terms, MND yields a MAE of approximately 94.5, whereas MND+GPS reduces this to about 50.5. Similarly, the relative error (MAPE) for MND is around 129.6%, compared to 40.6% for MND+GPS. These results confirm that MND on its own substantially overestimates satellite values, while MND+GPS provides a more proportionate adjustment. Both methods correlate well with the satellite data ($r \approx 0.82$, as shown in Table IX), indicating that they capture temporal and spatial trends, but the magnitude of uncorrected MND values is excessively high. Incorporating GPS not only improves accuracy but also significantly reduces the degree of overestimation.

VI. DISCUSSION

A. Results Discussion

The validation process, based on the direct comparison between presence indicators and satellite-derived person counts, shows a reasonable level of correlation with the MND+GPS-based presence estimates (Table VIII).

However, the comparison is subject to inherent interpretative constraints due to the differing nature of the data sources. The validation relies on satellite imagery, which provides high-precision but instantaneous and weather-dependent snapshots. Unlike MND+GPS, which captures continuous dynamic flows, the satellite “ground truth” is limited to specific timestamps and is susceptible to undercounting in areas with heavy shadow occlusion or visual obstruction. Consequently, the validation should be interpreted as a confirmation of the model’s spatial distribution capabilities under clear-sky conditions, rather than a full temporal validation of dynamic population fluxes. Discrepancies between the two metrics must therefore be viewed through the lens of these observational limits, where satellite counts often represent a conservative “lower bound” of the actual outdoor presence.

The observed discrepancies in specific cases can be partially explained by the different sensitivities of the data sources. High satellite values on particular days may reflect unusual crowd concentrations that were effectively captured by the satellite imagery but not by the GPS probability maps. While MND is capable of detecting sudden and localized increases in population density, the GPS probability maps are constructed by aggregating data over several months, which smooths out short-term fluctuations and reduces temporal granularity. This methodological difference explains why MND alone can register extreme peaks that MND+GPS does not reproduce with the same intensity.

Additionally, the systematically high values observed in Sol with MND may be attributed to the structural characteristics of the area. Sol is a major transportation hub, integrating both metro and commuter rail stations. Consequently, MND and GPS technologies, which can detect the presence of individuals even when located underground, register elevated values. In contrast, satellite imagery is limited to capturing the population present on the surface. This intrinsic limitation of the satellite benchmark contributes to the persistent overestimation observed in MND at this location.

The interpretation of results must, however, consider key sources of bias. On the one hand, the typology of mobile applications generating GPS traces is not completely transparent to the researchers, which may lead to an overrepresentation of certain behaviors, such as outdoor movements captured by navigation or fitness apps, potentially skewing the estimated probability of presence.

On the other hand, Earth Observation (EO) data introduces several limitations when used to infer outdoor population presence. A key limitation lies in the domain gap between the training data (orthophotos from the Madrid Geoportal) and the target satellite imagery. Although the orthophotos were downsampled

to 15 cm to match Pléiades Neo resolution, differences in acquisition geometry (e.g., off-nadir angle), radiometry, and platform characteristics can still affect generalization, particularly when applying the model to new sensors such as Maxar. While a light fine-tuning step was performed using Pléiades Neo imagery, this adaptation was limited and may not fully bridge the domain shift, particularly when applying the model to images from other sensors such as Maxar. This sensor dependence was reflected in the lower accuracy observed on the Maxar scene compared to the Pléiades Neo acquisitions. Differences in sun angle, contrast, and shadow sharpness across sensors likely contribute to these performance gaps.

Another challenge stems from the reliance on shadow-based detection: the presence of a pedestrian is inferred indirectly from the shadow they cast on the ground. As such, when individuals are located in shaded areas—due to building occlusions, trees, or overhangs—they may go undetected. This leads to a systematic underestimation of presence levels, with the degree of bias depending heavily on urban morphology (e.g., street width, building height) and seasonal conditions (e.g., sun angle and shadow length). In densely built environments or narrow streets, these occlusions can drastically reduce detectability. Additionally, urban clutter—including street furniture, lampposts, vehicles, and rooftop elements—can cast shadows that mimic those of pedestrians, resulting in false positives. These artifacts are particularly common in high-resolution imagery where such small features are clearly visible, yet difficult to disambiguate based on geometry alone. A further constraint of EO-based monitoring lies in its limited temporal resolution. Most satellite acquisitions represent a single snapshot in time, often around local solar noon. As a result, EO imagery cannot capture population dynamics across the day (e.g., morning vs. afternoon peaks), making it difficult to fully characterize temporal fluctuations in outdoor activity. This restricts the granularity of validation and reduces the ability to compare with ground-based, time-resolved data sources such as mobile network activity. Finally, the effectiveness of shadow-based pedestrian detection is highly dependent on sensor characteristics, including viewing geometry, radiometric quality, and acquisition angle. As observed in this study, detection performance varied significantly between images acquired by Pléiades Neo and Maxar, despite having similar spatial resolution (15 cm). Differences in solar elevation, sensor viewing angle, and radiometric contrast can substantially alter the visibility and sharpness of shadows, thereby affecting both false positive rates and missed detections. This sensor sensitivity must be considered when comparing EO-derived presence estimates across platforms or time periods.

B. Future Outlook

Further research is still needed on how we can obtain and clean geospatial data from personal devices. New IoT technologies that connect mobiles and wearables with different infrastructures and vehicles in a smart city context will add information to investigate how citizens make use of public spaces.

In addition, obtaining real-time population data can be challenging due to privacy and computational constraints. In this

scenario, the use of historical data combined with predictive modeling becomes essential for reliable, time-sensitive population estimation. Leveraging MND alongside contextual variables, such as day of the week, time of day, weather conditions, or holiday periods, enables the development of machine learning models capable of producing accurate near real-time population estimates.

Validation with satellite images provides a reasonable level of agreement between both sources. However, the temporal limitation of this technology prevents us from addressing the validity of the continuous estimation of the population. Other data sources may be used in further research to validate the monitoring methodologies, such as: images and videos captured by drones flying above the cities; stadium and concert attendance estimates; fixed sensor networks such as people counting cameras, public space occupancy sensors; beacons and Bluetooth/WiFi sniffers, which detect mobile devices to estimate density and movement without identifying the user and are used in shopping malls, stations, airports, etc.; and leveraging the 5G infrastructure, which allows for a more precise location due to their small cells that allow users to be located within a few meters and technologies such as beamforming and Multiple Input Multiple Output (MIMO) that help fine-tune location.

VII. CONCLUSION

The validation results demonstrate that MND enhanced with GPS provides a reasonable approximation of population presence in urban open spaces, especially when compared with satellite-derived person counts. The degree of agreement observed in key public squares of Madrid supports the feasibility of using MND+GPS as a viable method for continuous and scalable population monitoring.

The continuous monitoring of the population is key to understanding the impact of new technologies, e.g., Innovative Air Mobility, Cooperative Connected and Automated Mobility (CCAM); how urban changes will affect population; how people are exposed to noise and other pollutants; crowd management; or event planning. The massive use of mobile personal devices and the access to MND and GPS data make it possible. Despite the limitations, the use of MND+GPS offers a distinct advantage for capturing temporal variations in population distribution, which EO imagery cannot provide due to its single-time-point nature. This makes MND+GPS particularly suitable for dynamic urban analyses and time-sensitive applications.

As cities evolve into complex, data-driven ecosystems, understanding how populations interact with space and time will become foundational to designing equitable, sustainable, and resilient urban environments. The fusion of mobile network intelligence, pervasive sensors, and emerging communication technologies opens a pathway toward real-time, adaptive models of human presence. This vision calls for interdisciplinary collaboration, robust data governance, and a commitment to ethical innovation, where population mapping becomes not just a technical tool but a catalyst for shaping cities that are more responsive to the needs, behaviors, and rhythms of their citizens.

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